Vedant Sathye

669.225.3470 | vedant.sathye@gmail.com | https://www.linkedin.com/in/vsathye

WORK EXPERIENCE

ActZero
Machine Learning Engineer
August 2023 - Present

- Productionized data pipelines to train variational autoencoders for anomaly detection on over 2000 endpoint behaviors and serve at 2x the previous rate
- Continuously populated a neo4j graph database to visualize and track endpoint networking behavior and their relational properties
- Deployed RAG and prompt engineering methodologies for efficient one shot extraction of salient information from knowledge graphs
- Created and improved agentic workflows improve efficiency of cybersecurity research by 300%
- Designed pipelines to A/B test LLM fine-turning deployments and utilized Langfuse to monitor cost and performance

Valence Vibrations
Lead Machine Learning Engineer
October 2022 - August 2023

- Utilized LSTMs and voice analytics to extract 7 different emotional undertones from human speech
- Downsized trained model for efficient runtime on embedded systems while maintaining accuracy within 5% of original
- Deployed production ML model API on AWS Lambda

Oracle
Machine Learning Intern
June 2022 - September 2022

- Created pipeline to parse and transfer 2TB of SEC EDGAR filings to Open Search cluster for ease of querying during model training
- Trained 50k feature tokenizer and pre-trained BERT Base model on EDGAR data
- Trained and improved Named Entity Recognition model to extract causes and effects from financial text with 60% annotation level accuracy
- Utilized feature attribution library to interpret ML model behavior

Schrödinger
Machine Learning Intern
June 2021 - September 2021

- Integrated and improved on a novel Reinforcement Learning model to learn properties of graph-based molecules
- Optimized ML models on an AWS cloud environment
- Increased accuracy of results by 50% from baseline

Meyer Laboratory
Machine Learning Researcher
March 2020 - June 2021

- Created and trained Parafac and Tucker Decomposition models to observe patient protein data with a reconstruction success rate of 95%
- Utilized Gene Set Enrichment Analysis to observe efficacy of hormone treatments on MRSA patients

RelayPlay
Machine Learning Intern
April 2020 - December 2020

- Designed and tested a Natural Language Processing Pipeline to turn event descriptions into categorical features
- Created a Machine Learning Model and paired it with a stable matching algorithm to match user interest with preferred social events

EDUCATION

University of California Los Angeles | Los Angeles — **Class of 2023**
MS Computer Science Specialization in ML&AI

University of California Los Angeles | Los Angeles — **Class of 2022**
BS Computer Science | 3.70 GPA

SKILLS

- Python
- AWS
- ML/AI
- Model optimization
- Sci-kit Learn
- Keras
- Pytorch
- Graph DBs
- Pandas
- Serverless
- LLMs/Agents
- VectorDBS
- Tensorflow
- Git
- Bash
- Neo4j
- SQL
- NLP
- Docker

Publications

- Shah, Nilay S., et al. "Cardiovascular and Cerebrovascular Disease Mortality in Asian American Subgroups." *Circulation: Cardiovascular Quality and Outcomes* (2022): 10-1161.
- Huang, Robert J., et al. "Disaggregated Mortality from Gastrointestinal Cancers in Asian Americans: Analysis of United States Death Records." *International Journal of Cancer*, vol. 148, no. 12, 2021, pp. 2954–2963., doi:10.1002/ijc.33490.